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A recurrent neural network approach to predicting hemoglobin trajectories in patients with End-Stage Renal Disease

Benjamin Lobo[a,*], Emaad Abdel-Rahman[c], Donald Brown[a], Lori Dunn[b], Brendan Bowman[c]

[a] Department of Systems & Information Engineering, University of Virginia, Charlottesville, VA 22904, United States
[b] Medical Center – Pharmacy, University of Virginia, Charlottesville, VA 22908, United States
[c] Division of Nephrology, Department of Medicine, University of Virginia, Charlottesville, VA 22908, United States

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ABSTRACT

The most severe form of kidney disease, End-Stage Renal Disease (ESRD) is treated with various forms of dialysis – artificial blood cleansing. Dialysis patients suffer many health burdens including high mortality and hospitalization rates, and symptomatic anemia: a low red blood cell count as indicated by a low hemoglobin (Hgb) level. ESRD-induced anemia is treated, with variable patient response, by erythropoiesis stimulating agents (ESAs): expensive injectable medications typically administered during dialysis sessions. The dosing protocol is typically a population level protocol based on original clinical trials, the use of which often results in Hgb cycling. This cycling phenomenon occurs primarily due to the mismatch in the time between dosing decisions and the time it takes for the effects of a dosing change to be fully realized. In this paper we develop a recurrent neural network approach that uses historic data together with future ESA and iron dosing data to predict the 1, 2, and 3 month Hgb levels of patients with ESRD-induced anemia. The results of extensive experimentation indicate that this approach generates predictions that are clinically relevant: the mean absolute error of the predictions is comparable to estimates of the intra-individual variability of the laboratory test for Hgb.

1. Introduction

Patients with End-Stage Renal Disease (ESRD) suffer from a variety of ESRD-induced conditions, with one of the major ones being ESRD-induced anemia. One of the main causes of ESRD-induced anemia is a non- or limited-functioning kidney that lacks the ability to produce erythropoesis stimulating agents (ESAs) which stimulate the production of red blood cells in the bone marrow in the presence of the required raw materials (e.g., iron). In order to address the ESRD-induced anemia, patients on dialysis (hemodialysis or peritoneal dialysis) are given ESAs as part of their dialysis regimen. Several studies have demonstrated that the optimum target range for hemoglobin in patients with ESRD on dialysis is 10–12 g/dL [1]. Determining the correct dose of the ESA that will bring the patient's hemoglobin (Hgb) level into the target range is a complicated process because the correct dose varies not only from one patient to another, but can vary in the same patient over time depending on a number of patient specific factors (e.g., inflammation).

A primary issue with the protocols used for ESA dosing is that, while it takes 2 to 3 months [2] for the effect of a dosing change to be fully realized, the time between dosing decisions is typically 1 month. In other words, if a physician makes a dosing change now, the desired effect of this dosing change will not be reflected in the patient's Hgb level until 2 or 3 months later; *however*, the protocol has the physician check the patient's Hgb level just 1 month later at which point another change to the weekly ESA dose can be made. This mismatch invariably results in the physician being faced by one of the following situations:

1. One month ago the patient's Hgb level was below the target range and the dosing decision made was to increase the ESA dose. The patient's current Hgb level is even lower than it was one month ago; or
2. One month ago the patient's Hgb level was above the target range and the dosing decision made was to decrease the ESA dose. The patient's current Hgb level is even higher than it was one month ago.

In either situation the physician or anemia manager must choose between following the protocol and changing the weekly ESA dose again (increasing or decreasing it respectively), or going against protocol and keeping the weekly ESA dose the same. Invariably the decision, driven by the protocol, is to change the weekly dose to try and achieve the desired effect on the patient's Hgb level by the following month.

This pattern of changing the weekly ESA dose will continue until the patient's Hgb rises or falls into range. However, by this time the

* Corresponding author.

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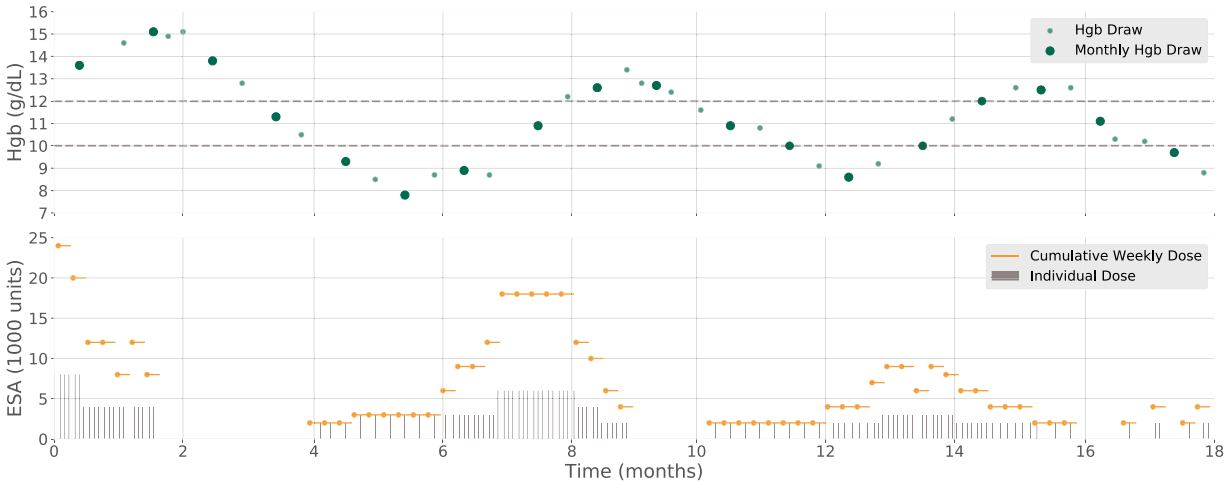

Fig. 1. An example of a Hgb time series that illustrates hemoglobin cycling. The Hgb time series (top plot) is sinusoidal in nature with the peaks and troughs falling outside of the desired target range (10 to 12 g/dL, indicated by horizontal dashed lines). The ESA dosing time series (bottom plot) shows the individual (vertical bars) and cumulative weekly (horizontal lines) ESA dose for the patient. The data shown in this figure is anonymized patient data.

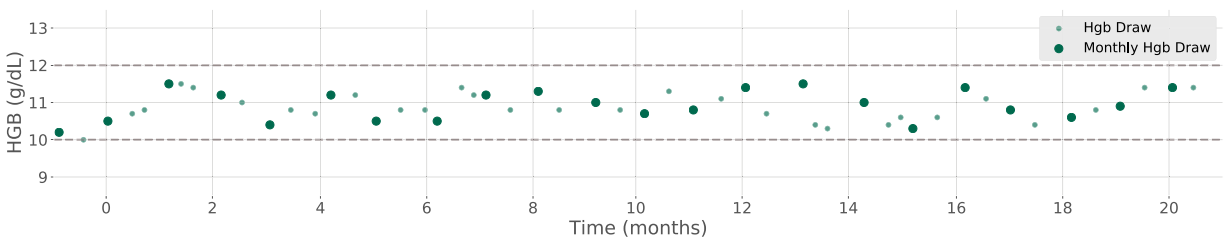

Fig. 2. An example of a Hgb time series that illustrates the desired long-term Hgb behaviour: there is still a sinusoidal component to the Hgb time series, but the peaks and troughs are contained inside the target range (10 to 12 g/dL, indicated by horizontal dashed lines). The data shown in this figure is anonymized patient data.

patient's weekly ESA dose is such that the patient's Hgb level will continue to rise or fall until it is again outside of range, but on the *other side*; and this situation will again trigger the dosing behavior that lead to the Hgb level being out of range most recently. And thus this pattern of dosing, driven by the protocol that is typically used, produces the phenomenon of hemoglobin cycling [3]. The Hgb time series shown in Fig. 1 is a prototypical example of hemoglobin cycling: the Hgb time series is sinusoidal with peaks and troughs that lie outside of the desired target range (indicated by the horizontal dashed lines at 10 g/dL and 12 g/dL). In addition, the time series of the cumulative weekly ESA dose shown in Fig. 1 clearly shows the role that the pattern of ESA dosing has played: the shape of the cumulative weekly ESA dose time series is sinusoidal in nature, just like the Hgb time series, but is shifted to the left by approximately 2 months, indicative of the time lag (delay) between a change in weekly ESA dose and the effect of that dose change being reflected in the patient's Hgb level.

Because the protocol is the standard of care, the physician or anemia manager must have a compelling reason to deviate from the protocol. This need for a compelling reason forms a psychological barrier that makes it is easier for the physician or anemia manager to do “something” (follow the protocol and change the weekly ESA dose) than to do “nothing” (go against protocol by keeping the weekly ESA dose the same, but come up with a compelling reason for doing so). If the physician or anemia manager has reason to believe that not changing the weekly ESA dose this month will result in the patient's Hgb level beginning to increase (or decrease depending on the situation) 1–2 months from now, because 1–2 months from now the effect of the weekly ESA dose change made 1 month ago from today will have had enough time to be fully realized, then they might feel comfortable with not making a change to the weekly ESA dose despite the fact that the patient's Hgb level will continue to decrease (respectively increase) in the short term.

A precise and accurate prediction of the patient's Hgb level 1, 2, and even 3 months into the future (a hemoglobin trajectory) would provide the physician or anemia manager with the information needed to deviate from the protocol and keep the weekly ESA dose the same even though the patient's Hgb level is still out of range. For example, if the 1 month prediction indicates that the patient's Hgb level will continue to drop, but the 2 and 3 month predictions indicate that the patient's Hgb level will begin to rise *despite* the weekly ESA dose being kept the same over the next three months, then the decision maker might be comfortable with making no changes. As a result, hemoglobin cycling should be drastically curtailed or even eliminated because dosing changes will be made only *after* the full effect of the most recent dosing change has been realized. Note that even in the absence of hemoglobin cycling there will still be month-to-month variation in the Hgb level similar to what is seen in Fig. 2: there is still a sinusoidal component to the Hgb time series, but the peaks and troughs are contained inside the target range.

Patient response to ESAs is variable and is, in part, a function of the availability of the iron required to build red blood cells. As such, a critical part of successfully managing anemia in patients with ESRD-induced anemia is the managing the combination of both ESA dosing and iron administration [4]. This research presents a recurrent neural network approach to predicting Hgb trajectories in patients with ESRD-induced anemia. The predicted Hgb levels 1, 2, and 3 months from the current point in time are conditioned on a 1, 2, and 3 month ESA *and* iron dosing strategy. ESA dosing and iron dosing are considered simultaneously so that their combined effect on Hgb levels can be modeled and accounted for when generating Hgb predictions. Section 2 highlights the previous work done in this area and outlines the contributions of this research. Section 3 goes over the data used for modeling while Section 4 provides the model details. Section 5 presents the results of an extensive design of experiments to evaluate the modeling

approach, along with discussion of the results. Finally Section 6 presents the conclusions and some ideas for further research.

2. Literature review

Predicting future Hgb levels for patients with ESRD has been an active area of research for more than two decades. During this time a wide variety of modeling approaches have been used to address the problem, ranging from linear regression to control theory models to a large number of machine learning techniques including self organizing maps, random forests, and artificial neural networks. Although the relevant literature is focused on predicting future Hgb levels for patients being given ESAs to address their ESRD-induced anemia, there are a number of different ESAs that can be used to treat the anemia. Each type of ESA has different pharmacokinetics and pharmacodynamics (see for example [5–7]), and these differences mean that a model that predicts Hgb levels for patients using one type of ESA will not work well, or perhaps not work at all, for patients using a different ESA. For example, Darbepoetin alfa is a second generation ESA that is chemically different from Epoetin (EPO) and engineered to provide reduced clearance from the blood. This yields a significantly increased half life (approximately three times that of EPO) and longer time of action per given dose in comparison to EPO. Dosing frequency is reduced as well with Darbepoetin typically administered at intervals of once weekly, every two weeks, or even monthly. Brier et al. [8] contains an overview of the work done to model (predict) hemoglobin response to ESAs, and what follows is a review of the most relevant literature to the problem being addressed in this work.

Martín-Guerrero et al. [9] investigated the use of Profile-Dependent Support Vector Machine (PD-SVM) models to forecast (predict) Hgb levels. They compared the PD-SVM models with the Multilayer Perceptron (MLP) model as well as the Autoregressive Conditional Heteroskedasticity model, a model in which error variance changes over time. They performed one-step ahead (one month ahead) prediction on two types of ESAs (Epoetin alfa and Epoetin beta), and found that the PD-SVM model performed best, but was only very slightly better than the MLP model. A limitation of the work was that the cohort used was small and homogeneous in nature.

Soria et al. [10] used Self-Organizing Maps (SOM) to extract qualitative information from the patient data, and then used a MLP model to predict the Hgb level one month on using information from the current and previous months. Both Epoetin beta and Darbepoetin were used as the ESA. The SOM were used to find relationships between Hgb level and the other features used (weight, gender, age, previous Hgb level, ESA and Fe in the preceding two months). The SOM indicated that there were relationships between (a) low doses of ESA and good levels of Hgb level in younger male patients, and (b) low weight females and elderly patients who were stable (no large changes in Hgb level). The MLP model predicted a Hgb level one month ahead using weight, gender, and age, and the Hgb level and ESA dosing information from the current and previous months. They provided the Root Mean Squared Error, ME, and MAE for the differences between observed and predicted Hgb levels, and compared linear regression models fitted with robust least-squares to the MLP model. They found that the two models perform similarly for Epoetin beta, while the MLP models performed better for Darbepoetin. They theorized that excluding outlier patients (those with multiple co-morbidities and other special situations) would boost predictive performance of the models.

Martínez-Martínez et al. [11] used a machine learning model to predict the next month's Hgb level using global models (a single model for an entire cohort of patients) and local models (different models for different patient profiles). They used Adaptive Resonance Theory and hierarchical clustering to cluster snippets of patient histories. The clustering is based on 4 kinds of data:

1. The current and four previous Hgb levels,
2. The current and four previous ESA dose values,
3. The current and time derivatives of the differences between consecutive Hgb levels in the four previous samples, and
4. The current and time derivatives of the differences between consecutive Hgb levels in the four previous samples while including the ESA dose values.

The clusters were then used when building what were termed local models – models built using only data from a particular cluster. Although a large number of statistical and machine learning models were considered (linear regression, random forests, MLP models, SVM models), there was no statistically significant difference in the performance of the best local and best global models (using the Root Mean Squared Error and MAE performance metrics).

Akabua et al. [12] used semi-blind robust identification (a technique that is based on feedback control theory) to identify individual patient models of anemia using the Hgb and ESA dosing information. They built a base model and then personalized the model using the individual patient's history. Their technique novelty stemmed from being able to predict based on very limited amounts of data (minimal historic information) and without using any demographic information. However they only provided results on 5 selected patients, and did not provide summary statistics on the 50 patients that they had in their cohort, which made it difficult to assess the true merit of the approach.

Barbieri et al. [13] recognized that predicting future Hgb levels "is intrinsically a temporal problem" and wanted to encode the temporal structure of the data in the modeling process. The ESA used was Darbepoetin alfa. They had a very large cohort of patients from three different countries (Spain, Portugal, and Italy), and built MLP models using this data. The results presented were an improvement on both the linear models built using the same data and the MLP model built by Martínez-Martínez et al. [11].

Barbieri et al. [14] used an Artificial Neural Network (ANN) model to predict Hgb levels 3 months into the future to "support decision-making regarding anemia management in hemodialysis patients" where the ESA was intravenous Darbepoetin alfa. Their model was built to take into account the drug kinetics of the ESA being used and the life cycle of red blood cells, which is why they chose to predict Hgb levels 3 months into the future. They observed that their model performance could be improved upon by removing events (such as hospitalizations or transfusions) that were the main causes of large errors and discrepancies in the predictions.

Fuertinger et al. [15] present a method to predict hemoglobin levels in individual patients using a mathematical model of erythropoesis. One limitation of their work is that the results are based on just 60 patients, which makes it difficult to understand how these results would apply to the population in general. However, the main limitation of this approach is that it requires specific monitors that (non-invasively) measure hemoglobin on a per-treatment level, but these monitors are not the standard of care, and thus the data required to build the model is not available in most dialysis settings.

All of the literature discussed above only models future ESA dosing; iron dosing, which is a key component of anemia management in dialysis patients [4], is not considered. The main contribution of the work presented in this paper is a model that uses future ESA dosing *and* iron dosing information to predict the Hgb levels of dialysis patients with ESRD-induce anemia. To our knowledge this is the first work that considers future ESA dosing and iron dosing simultaneously [8]. In addition, all of the literature discussed above uses a fixed forecasting horizon. For example, in the paper by Barbieri et al. [14], a 3 month forecasting horizon is chosen. The issue with this method is that when the physician or anemia manager is looking at the next 3 months of Hgb level predictions to assess the Hgb level trajectory, the prediction (forecast) 1 month in the future is already 2 months old and the prediction 2 months into the future is already 1 month old. Clinical events like a hospitalization or other life events that have occurred since those

predictions were made might render them obsolete. In contrast, a system in which the Hgb level predictions for 1, 2, and 3 months into the future are generated each month will always make use of the most up to date information when making those predictions, and thus will provide the most accurate predictions for each forecasting horizon and for the Hgb trajectory as a whole. Such a system is the approach taken in this research which, each month, generates predictions of the Hgb level 1, 2, and 3 months into the future. At any given month, the 1, 2, and 3 month predictions made are the best estimate of the patient's Hgb levels 1, 2, and 3 months into the future, and provide the best estimate of *how* the Hgb level arrives at the 3 month prediction over the next 3 months. And this Hgb trajectory is of critical importance to decision makers like physicians or anemia managers who must stand by dosing decisions that may seem counter-intuitive (and go against protocol) given the patient's current information.

3. Data

3.1. General observations

The data set encompasses the dialysis, dosing, and laboratory test history of 1972 patients that came to one or more of the 11 dialysis clinics that form the University of Virginia (UVa) dialysis system (see Fig. 3), during the time period starting on January 2nd 2009 and ending on December 31st 2018. Each patient was at least 18 years of age at the time of their first dialysis visit, dosing event, or laboratory test order. The data was obtained under UVA Health Sciences Research IRB protocol number 20959. In Fig. 4, the number above each bar indicates the number of unique patients who have at least one data point (dialysis, dosing, or laboratory test data) in each year, while the bar indicates the gender split of the patient population by year. While the dialysis system population has grown over time the gender split has remained relatively constant. Each patient has a different amount of data; Fig. 5 shows the distribution of the number of monthly Hgb values per patient, and shows that some patients belonged to the UVa dialysis system for over 5 years while others belonged to the system for a few months. This is of note because it means that a particular patient might not have enough data for the modeling process. For example, if 2 months worth of historic Hgb data is required as input to the modeling process, then to predict a Hgb level 1 month in the future requires 4 consecutive months of data – 2 months of historic Hgb data, the Hgb data from the current month, and the Hgb data from the month being forecast. And if a patient has 5 months of historic Hgb data, say January through the end of March, and then June and July, their data would not be a part of the modeling process because they do not have at least 4 months of consecutive Hgb data, even though they have 5 months worth of historic Hgb data.

Table 1 displays the summary statistics of the entire patient cohort for select measures.

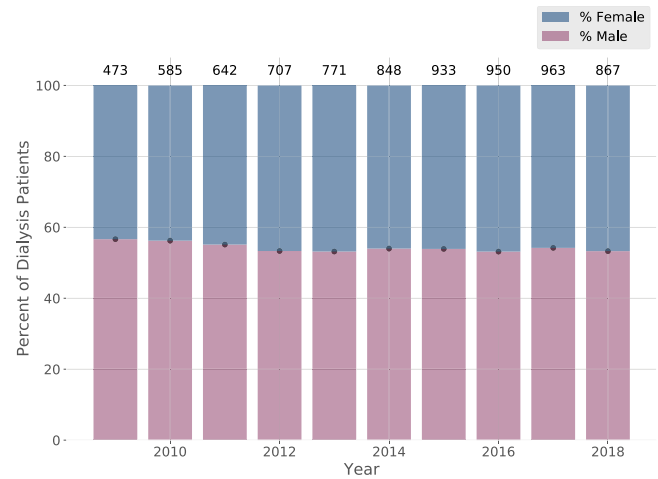

Fig. 4. Patient population and gender split by year. The number above each bar indicates the number of unique patients who have at least one data point (dialysis, dosing, or laboratory test data) in that year.

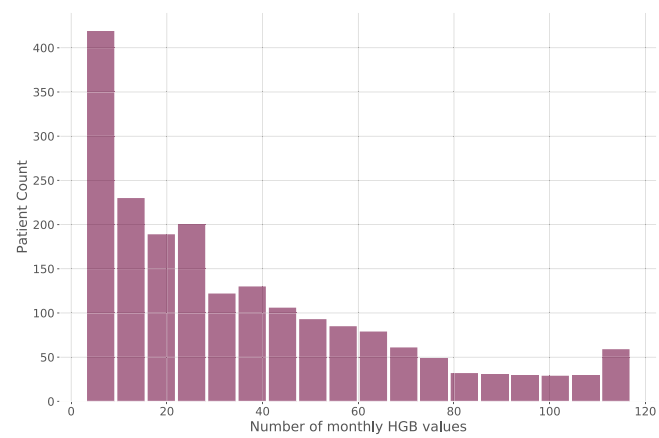

Fig. 5. Distribution of the number of monthly Hgb values per patient. Patients with more than 60 monthly Hgb draws have been part of the UVa dialysis system for over 5 years.

3.2. Hgb draw data

Each patient in the UVa dialysis system is required to have a monthly Hgb draw that occurs at some point during the first full two weeks of the month, thus every patient has *at least* one Hgb data point each month that they are on dialysis. The time between Hgb draws is dictated by Centers for Medicare & Medicaid Services (CMS) guidelines and are consistent across all dialysis organizations. Physiologically speaking, the 1 month lab interval is the typical interval used nationally as it represents the shortest possible time after which it is possible to

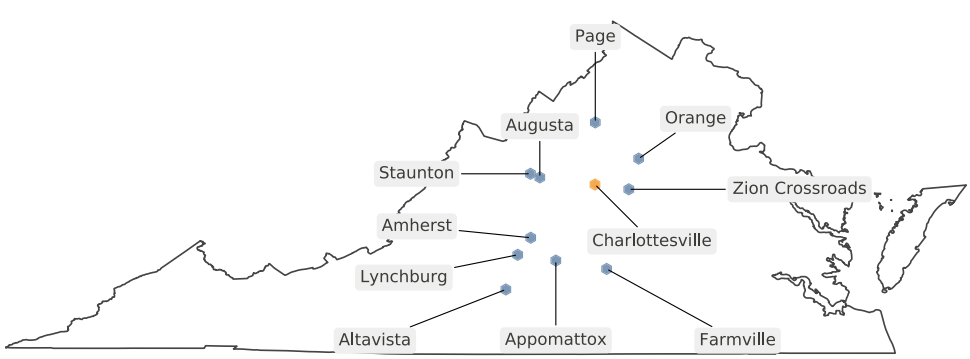

Fig. 3. The geographic locations of the 11 dialysis clinics across Virginia that form the UVa dialysis system.

Table 1
Patient cohort summary statistics for selected variables.

Measure	Mean (standard deviation)
Patient Characteristics	
Time on dialysis (weeks)	157.62 (134.78)
Age at first visit (years)	61.44 (14.65)
Pre-dialysis weight (lb)	87.76 (25.75)
Post-dialysis weight (lb)	85.42 (25.28)
Dialysis	
Dialysis session duration (minutes)	212.16 (36.45)
Number of dialysis sessions per week	2.68 (0.83)
Dosing	
EPO dose (units)	5340.41 (5011.77)
IV Iron dose (mg)	95.51 (35.89)
Laboratory Tests	
Albumin (g/L)	3.65 (0.41)
Calcium (mg/DL)	8.74 (0.79)
Carbon Dioxide (mmol/L)	23.60 (3.11)
Creatinine (mg/dL)	8.60 (3.32)
Sedimentation Rate (mm/h)	47.48 (30.82)
Ferritin (ng/mL)	899.34 (551.92)
Folate, Serum (ng/mL)	10.82 (5.21)
Glomular Fill Rate (mL/min)	6.73 (4.00)
Hemoglobin A1c (%)	6.72 (1.53)
Hematocrit (%)	33.30 (4.23)
Hemoglobin (g/dL)	10.63 (1.38)
Iron (mcg/dL)	60.66 (27.29)
Potassium (mmol/L)	4.64 (0.73)
Kt\ V	1.53 (0.29)
Mean Corpuscular Volume (fL)	94.90 (6.92)
Sodium (mmol/L)	137.95 (3.49)
Protein Catabolic Rate (g/kg/day)	0.97 (0.28)
Phosphorus (mmol/L)	5.45 (1.65)
Percent Saturation (%)	26.34 (11.69)
Parathyroid Hormone Intact (pg/mL)	580.14 (451.10)
Int. Normalized Ratio	2.13 (0.98)
Red Cell Distribution Width (%)	14.93 (1.60)
Transferrin (mg/dL)	164.20 (34.96)
Urea Reduction Ratio (%)	72.64 (6.71)
Vancomycin LV (ug/mL)	16.64 (6.94)

observe a change in Hgb level that is the result of a change in the weekly ESA dose. In the current regulatory environment, dialysis organizations bear the cost of Hgb draws and, absent compelling clinical indications, the monthly interval is thought sufficient for clinical care. Some patients will have additional Hgb draws during the month for a number of reasons, including:

1. If their ESA dose was changed following their monthly Hgb draw. In this case by protocol they get a Hgb draw approximately two weeks later,
2. If there is reason to suspect that there was something wrong with the most recent Hgb draw (e.g., the Hgb level came back unusually low or high compared to the last Hgb draw),
3. If their Hgb level lies outside the target range of 10–12 g/dL. In this case the protocol dictates how often to redo the Hgb draw; or
4. If there are clinical indications (e.g., suspected gastrointestinal bleeding).

Examples of the first reason can be seen in Fig. 6, where there are Hgb draws in weeks 3, 7, and 12 which occur approximately midway between the monthly draws which occur in weeks 1, 5, 10, and 14. The weekly Hgb draws in weeks 16, 17, and 19 are an example of more frequent Hgb draws occurring when the patient's Hgb level lies outside of the desired range. Because the number of Hgb draws varies between patients, not every patient has the same Hgb data density, and the modeling approach should take this into account.

The number of weeks in between the monthly Hgb draws can also vary. This occurs mainly due to the fact that the number of weeks in a given month varies, but can also be due to a patient missing dialysis on their regularly scheduled draw date and having their Hgb draw occur at a later date. Fig. 6 also provides an example of this, where the time between the first and second monthly Hgb draws shown is 4 weeks, while the time between the second and third monthly Hgb draws is 5 weeks.

3.3. Laboratory tests data

Each patient has a variety of laboratory tests that can be ordered. The original data set as pulled from the dialysis database included instances of test values incompatible with life, i.e., erroneous test values. In order to clean the data, we defined thresholds for each laboratory test that in our judgement would balance the goal of excluding test values that were clearly errors while still including legitimate test values that happen to be outliers. As there is no mechanism to adjudicate a potential lab error versus a legitimate outlier, we elected to use more inclusive ranges for all laboratory tests to minimize the risk of censoring legitimate outliers. Thus the laboratory test ranges in Table A.1 of the appendix represent the qualitative clinical judgement of the physicians of extreme, but possible, test values. Cleaning of the laboratory test data was accomplished by removing a test value from the data set if the test value lay outside the range given in Table A.1 of the appendix for that particular test.

Note that the draw schedules for tests can be different; for example Iron and Percent Saturation levels are checked monthly while Ferritin is checked every three months, and the modeling approach should take this into account.

3.4. Dialysis data

The vast majority of patients receive dialysis three times a week, either Monday, Wednesday, and Friday (MWF), or Tuesday, Thursday, and Saturday (TTS). However, there are times when a patient fails to come in for dialysis (sick, hospitalized, or missed for other reasons), and times when a patient might come in for an extra session. Each time a patient comes in for dialysis there are a number of dialysis related measures that are captured. The ones used in the modeling process are the inter-dialysis weight, blood flow rate, ordered dialysis time, and actual dialysis time.

3.5. Dosing data

The UVa dialysis program utilizes recombinant human erythropoietin alpha (EPO), a 165-amino acid protein that is administered intravenously yielding a half-life of 6–8 hours in the body, as the ESA. The intravenous route is generally preferred as it is more convenient for dialysis patients than a subcutaneous injection. Because the data set is taken from a group of dialysis clinics across the state of Virginia (see Fig. 3), the dosing needs of each patient vary greatly. Most patients are dosed with EPO each time they come in for dialysis. However, there are also patients who are dosed only once or twice a week, and in rare cases only once every other week. This non-standard dosing makes modeling such a heterogeneous population that much more difficult, but the modeling approach presented here are able to handle such variation. ESA dosing is driven by protocol, where decisions are based on the current Hgb level and the change in Hgb level from last month. The protocol calls for a change in the weekly ESA dose if the current Hgb level lies outside the target range of 10–12 g/dL, and the change called for is based on the magnitude of the change in Hgb level when compared to the Hgb level 1 month ago. The weekly ESA dose is adjusted as a percentage of the current weekly ESA dose, so that if a patient currently receives 9000 units/week, a 25% increase results in a weekly ESA dose of 11,000 units per week (after the ESA minimum vial size of 2000 units is taken into account).

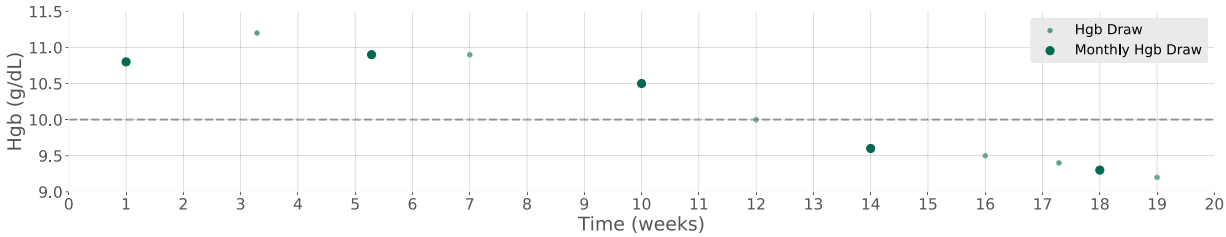

Fig. 6. An example of a Hgb time series that illustrates how both the number of weeks between monthly Hgb draws and the frequency of Hgb draws can vary over time. Circle locations indicate the Hgb level at draw time, and large circles indicate that the draw was a monthly draw. The data shown in this figure is anonymized patient data.

Iron dosing is also driven by protocol, with intravenous iron being preferred to oral iron as it ensures compliance and elminates the gastrointestinal side effects of oral iron. In the U.S. most dialysis organizations follow the general outlines of the "DRIVE" study [16]. The primary lab value utilized is the transferrin saturation (TSAT), a measure of the saturation of iron carrier molecules in the blood. In general, most iron dosing protocols recommend intravenous iron (at differing dosages) when the TSAT falls below 20% with a concomitant ferritin less than 1000 ng/mL. However, there is significant variability in physician agreement with iron protocols due to poor underlying clinical trial data and so iron is often given or withheld according to individual provider (physician) beliefs and preferences.

4. Modeling

The time series (sequential) nature of the data is the main driver when selecting the appropriate model to use. In particular, the fact that prior dosing decisions can impact the future observed hemoglobin level is a strong argument for using a model with the ability to remember (account for) these prior dosing decisions. The long short-term memory (LSTM) recurrent neural network (RNN) architecture is tailored to working with sequential and time series data, and the memory feature of this architecture distinguishes it from the multilayer perceptron model (MLP, a type of artificial neural network), which lacks this feature. These considerations motivate the use of LSTM RNN models, which are "among the most effective sequence models used in practical applications" [17,Chap. 10.10].

4.1. Model structure

The neural network model (see Fig. 7 for a graphical depiction of the network) has three distinct parts to it that handle the three distinct types of data that serve as input to the model:

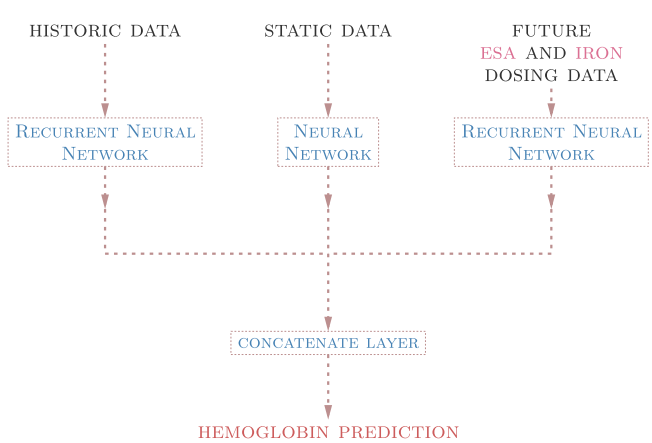

Fig. 7. Conceptual structure of the model illustrating the three main inputs (historic data, static data, and future dosing data), the modeling approach used for each data set, and the output (predicted Hgb level).

1. Historic data
2. Static data
3. Future dosing data

The historic data part of the network encompasses the history of the patient as a weekly time series, and is modeled as a RNN using `LSTM` layers in Keras[1] [18]. The input tensor of historic data moves through 4 `LSTM` layers and a `Dense` layer that are arranged in series. At a minimum the historic data includes the past Hgb draws and ESA dosing data, but can include the laboratory test data of the labs listed in Table A.1 of the appendix and the dialysis data of the dialysis measures listed in Section 3.4. The static data captures data which does not change or changes minimally over the historic data time frame (e.g., demographic data) and is modeled as a regular (vanilla) neural network using `Dense` layers in Keras. The input tensor of static data moves through 2 `Dense` layers that are arranged in series. The future dosing data part of the network encompasses the future weekly doses of ESA and iron over the forecasting horizon, and is also modeled as a RNN using `LSTM` layers in Keras. The input tensor of future dosing data moves through 3 `LSTM` layers and a `Dense` layer that are arranged in series. The output tensors from the final layer in each of the three network parts are brought together by a `Concatenate` layer in Keras (see Fig. 7), which then outputs the predicted Hgb level for the forecasting horizon.

Because the size of the input tensor to the RNN that models the historic data depends on the number of months of historic data being used, a separate model must be built each time the number of months of historic data being used is changed. And because the size of the input tensor to the RNN that models the future dosing data depends on the number of weeks in the forecasting horizon, a separate model must be built for each forecasting horizon under consideration. Thus in the results presented later a particular model is identified by the number of months of historic data used and the number of months in the forecasting horizon.

4.2. Building the input tensors

The model inputs described below were selected based on discussions with nephrologists and a close examination of the literature (for example see Table 1 in [13], Tables 1 and 3 in [14], and Box 1 in [19]). In order to build the input data tensors for the two RNNs using the data from a specific patient, each measure in the patient data first needed to be converted into a weekly time series. This was accomplished using the `resample` function in `pandas` [20].

Dosing: The ESA and iron time series were re-sampled using a weekly frequency, and the following measures were computed for the resulting time series:

1. The total dose administered that week,
2. The dosing count – the number of times they received a dose in that

[1] Details regarding any Keras function referenced in this paper can be found at https://keras.io/.

week, and
3. The dosing standard deviation – the standard deviation of all doses administered that week.

If no dose was administered or a metric could not be computed for a given week it was replaced with 0. In addition, a normalized weekly ESA time series was created by dividing the weekly ESA time series by the average patient weight during the week; the resulting weekly time series was then converted to a weekly time series representing the percentage change in dose week-on-week.

Hemoglobin: The Hgb time series was re-sampled using a weekly frequency, averaging Hgb levels if more than one Hgb level was measured in a single week. Because the model uses a weekly time series, ideally all patients would have a weekly Hgb draw to provide the most updated information to the model. However this is not the case. For weeks in which no Hgb was drawn, the model accounts for this by carrying forward the most recent Hgb level for up to 5 weeks (because the maximum time between Hgb draws should be 1 month), or by setting the Hgb level to 0 otherwise. In this way, the model makes use of the most up to date information on a patient.

Laboratory tests: For a given laboratory test, the laboratory test time series was re-sampled using a weekly frequency, averaging test values if more than one test value was ordered in a single week. The most recent test value was carried forward for a variable amount of time (depending on how often the test should be ordered), and otherwise was set to 0.

Dialysis: The dialysis time series was re-sampled using a weekly frequency, and the following measures were computed for the resulting time series:

1. The weekly dialysis count,
2. The mean of
   (a) The inter-dialysis weight,
   (b) The blood flow rate,
   (c) The actual dialysis time,
   (d) The patient weight at dialysis,
   (e) The patient's Body Mass Index (BMI), and
   (f) The patient's Body Surface Area (BSA).

Clinical events: In this context a clinical event is a hospitalization or a blood transfusion. A weekly time series was created that kept track of the number of days of that week spent in the hospital for hospitalizations, and that kept track of the number of units of blood transfused that week for transfusions.

Once the set of weekly time series was obtained for the specific patient the following algorithm logic was used to build samples for the input tensor. The algorithm looks for a data time window of length t where t is equal to the sum of the number of months of historic data plus the number of months of future dosing data. The algorithm then checks to make sure that:

1. There are $t + 1$ consecutive monthly Hgb draws in the data time window, and
2. There are no clinical events that occur *after* the current monthly Hgb draw but *before* the future monthly Hgb draw.

If the above criteria are satisfied, then the data during the time window can be used in the input tensors. This process was repeated for every patient, and the input data tensors were built sequentially in this fashion.

As an example, suppose the model is being built using 2 months of historic data and will be used to forecast the Hgb level 1 month out. In this case the data time window will be $t = 3$ consecutive months of data: 2 months of historic data and 1 month of future dosing data. This 3 month data time window must also include $t + 1 = 4$ consecutive monthly Hgb draws as illustrated in data time window A of Fig. 8: 2 historic monthly Hgb draws, the current monthly Hgb draw, and one monthly Hgb draw 1 month in the future. In data time window A, if there is a transfusion as indicated but no hospitalization as indicated, the data during time window A could still be used to generate a sample because the clinical event (transfusion) occurs in the "past" part of the time window. However, if in data time window A there is a hospitalization as indicated, the data during time window A could *not* be used to generate a sample because the clinical event occurs in the "future" part of the data time window. The data during the time window in data time window B in Fig. 8 *cannot* be used to generate a sample because, even though both clinical events occur in the "past" part of the time window, there is a missing monthly Hgb draw.

Each feature in the input tensor was normalized using the method suggested by Chollet [21, Chap. 3.6.2]: for each entry in a feature column, subtract the mean of the feature column and divide by the standard deviation of the feature column of the training data. The mean and standard deviation of each feature from the training data was used to normalize the validation and testing data sets.

4.3. General model details

The model was implemented in Python and built using Keras with TensorFlow [22] as the backend. The `RMSprop` optimizer was used with Mean Absolute Error as the loss function. The model was run using a batch size of 16 for up to 50 epochs, with an early stopping criterion implemented via the `EarlyStopping` callback in Keras.

In order to guard against information leakage, each patient was randomly assigned to one of the training set, validation set, or testing set. In this way, all of the input data generated from a specific patient went into a single, mutually exclusive, input data set. Because the amount of data required in a data time window is a function of the model being considered (i.e., the amount of data is a function of the number of months of historic data and the number of months of future dosing data), each patient generates different amounts of input data depending on the model under consideration. The average split in terms of samples in each set was 67.9% of samples in the training set, 17.4% of samples in the validation set, and 14.7% of samples in the testing set. The exact split for each combination of months of historic data and months of future dosing data (i.e., the forecasting horizon) is provided in Table A.2 of the appendix.

5. Results and discussion

A design of experiments was performed to determine the best model for each forecasting time horizon. The design took into consideration the following seven parameters:

1. Data used in the model: `Min` refers to using only the ESA dosing, iron dosing, and Hgb data to create the input tensors while `Full` refers to using all the patient data (lab tests, dialysis measures, etc.) available to create the input tensors.
2. ESA dosing normalization strategy: `PS` refers to creating the normalized weekly ESA time series by converting the weekly ESA time series to a time series representing the percentage change in dose week-on-week, while `WPS` refers to the normalized weekly ESA time series as described in Section 4.2.
3. Months of history: The number of months of historic data to use. This was one of 1, 2, 3, 4, or 5 months.
4. Network structure: `Orig` refers to using a `Dense` layer as the final layer to the RNNs (the historic data and future dosing data parts of the network), while `Mod` refers to using a `LSTM` layer as the final layer to the RNNs in the model (historic data and future dosing data).
5. Activation function: `None` versus `Relu` [23] indicates the type of activation function used in the `Dense` layers of the model.
6. Network size: `Big` refers to a network where the number of neurons in each layer are generally twice that of the number of neurons in

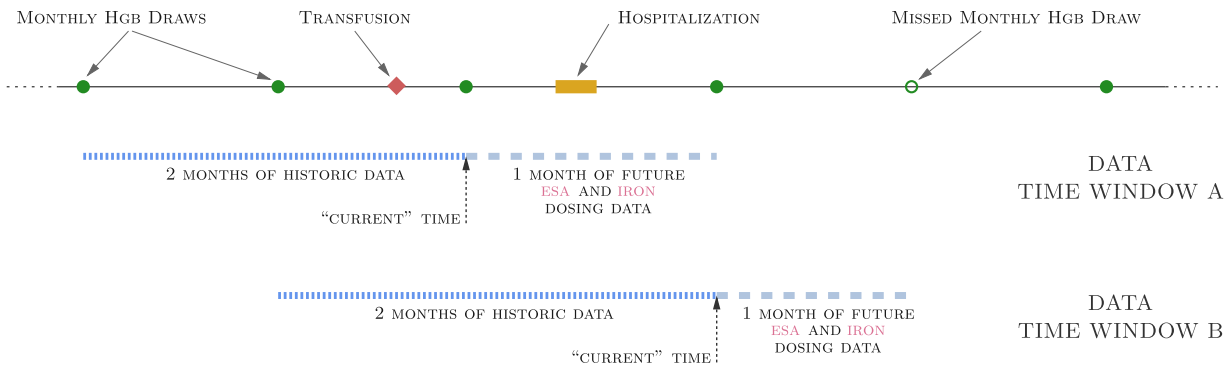

Fig. 8. Examples of data time windows for a model that uses 2 months of historic data and generates predictions/forecasts 1 month into the future.

each layer of an otherwise equivalent `Small` network. That is to say, if a `Small` network has 175 neurons per layer, then the `Big` network will generally have 350 neurons per layer if the other 6 parameters that define the `Small` network stay the same.

7. Dropout: `Med` refers to a network where the dropout and recurrent dropout (used to prevent overfitting) values are generally twice that of the dropout and recurrent dropout of an otherwise equivalent `Low` network. That is to say, if a `Med` network has a recurrent dropout of 0.1, then the `Low` network will generally have a recurrent dropout of 0.05 if the other 6 parameters that define the `Low` network stay the same.

There are 320 different combinations of the above 7 parameters, but because there are 3 different forecasting horizons (1, 2, and 3 months), there are actually 960 different models that were trained. The time needed to train a model ranged between 6 and 28 hours on a system with 18 cores (@2.3GHz) and 124GB RAM. The shortest times were taken by `Small` models which used `Min` data, 1 month of historic data, and were making a forecast just 1 month into the future. The longest times were taken by `Big` models which used `Full` data, 5 months of historic data, and were making a forecast 3 months into the future.

The following three metrics are commonly used in the literature to judge the performance of models used to predict future Hgb levels, and are defined here to facilitate understanding of the results that follow:

1. Mean Error (ME): Calculated as the simple average of the difference between the observed Hgb levels and the predicted Hgb levels. This statistic can be used to measure the bias of the model since the positive and negative errors tend to cancel each other out in an unbiased model.
2. Mean Absolute Error (MAE): Calculated as the average of the absolute difference between the observed Hgb levels and the predicted Hgb levels. This statistic is a scale-dependent accuracy measure (a MAE closer to zero is better).
3. Mean Squared Error (MSE): Calculated as the average of the squared difference between the observed Hgb levels and the predicted Hgb levels. This statistic is an accuracy measure as well as a measure of the quality of the model (a MSE closer to zero is better).

Both ME and MAE have units of g/dL since the Hgb level is measured in g/dL. Table A.3 of the appendix provides the performance metrics for the best performing model (see Table 2) on the training, validation, and testing sets for each of the 15 combinations of forecast horizon and number of months of history. The similarity in values of a performance metric (one of ME, MAE, and MSE) for a given model between the training and testing sets indicates that the models were *not* over trained. As such, the performance of the models on unseen data (i.e., new testing data) should closely mirror the performance on the training data.

5.1. General results

Table 2 lists the best performing model for each of the 15 combinations of forecast horizon and number of months of history. The best performing model is the one with the smallest MAE on the testing set (i.e., the smallest $MAE_{TESTING}$ value). The 'Network Configuration' column provides the details of the model (e.g., data used in the model, activation function, network size). The ME of the model on the training set is an indication of the bias of the model, and an indication of the likely bias that will exist when the model is run on the testing set. In order to account for the bias when using the model on the testing set, a modified version of each model was produced by adjusting the predicted Hgb levels for the testing set by the bias of the model on the training set (i.e., $ME_{TRAINING}$). A 'Yes' in the 'Model Modified?' column of Table 2 indicates that the performance metrics presented are for the $ME_{TRAINING}$ modified version of the model.

The results presented in Table 2 lead to the following observations:

- In general there is an incremental improvement in MAE to be had by increasing the number of months of history used.[2] This is expected because the model has more information to leverage as the number of months of history used are increased. Given that the life cycle of red blood cells is approximately 90 days, it might have been expected that there would not be a performance gain in using more than three months of historic data as that would be the limit of a change in the Hgb based on a change in ESA dosing. However, the improved performance that comes from using 4 and 5 months of history is an indication that there is predictive information encoded in the 4th and 5th month of historic data.
- For a given number of months of history, there is a decrease in performance the further into the future the forecast is (i.e., the MAE increases with an increase in the forecast horizon). This is also expected and in line with the general performance of predictive/forecasting models: the further into the future the prediction/forecast is, the greater the uncertainty surrounding the prediction/forecast. This phenomenon is visualized in Fig. 9, where the peak of the distribution of errors for the test set is lower and the tails of the distribution of errors for the test set are fatter as the forecast horizon increases. Fig. 9 also indicates the likelihood of different sizes of prediction error (i.e., the difference between observed and predicted Hgb levels). For the 1 month (2 month and 3 month respectively) predictions made using models built with 5 months of history, the absolute value of over 58% (53% and 52% respectively) of the prediction errors were less than 0.5 g/dL, the absolute value of over 86% (83% and 81% respectively) of the prediction errors were less

[2] It is not clear why the performance of the models that use 2 months of history do not present a performance improvement over the models that use just 1 month of history, nor why the 1 month of history models perform (marginally) better than the 2, 3, and 4 month(s) of history models.

Table 2
The best performing network configuration for each of the 15 different combinations of forecast horizon and months of history when future iron dosing information is used. The best performing network configuration for a given combination of forecast horizon and months of history is the network configuration that produces the minimum MAE value when used on the testing set.

Forecast horizon (months)	Months of history	Model modified?	Testing set			Network configuration
			ME	MAE	MSE	
1	1	Yes	0.010	0.541	0.521	[Min, PS, Mod, Relu, Small, Med]
	2	No	−0.008	0.548	0.535	[Min, PS, Orig, None, Big, Low]
	3	Yes	−0.004	0.537	0.513	[Min, PS, Mod, Relu, Big, Low]
	4	Yes	0.010	0.534	0.508	[Min, PS, Orig, Relu, Small, Low]
	5	Yes	0.017	0.527	0.489	[Min, WPS, Mod, Relu, Small, Med]
2	1	Yes	0.009	0.600	0.635	[Min, PS, Mod, None, Small, Low]
	2	Yes	0.027	0.608	0.669	[Min, PS, Mod, Relu, Small, Low]
	3	No	−0.046	0.605	0.664	[Min, PS, Mod, Relu, Small, Med]
	4	Yes	0.005	0.603	0.649	[Min, PS, Mod, Relu, Small, Low]
	5	No	−0.047	0.590	0.615	[Min, WPS, Mod, None, Small, Med]
3	1	No	−0.033	0.605	0.658	[Min, PS, Mod, Relu, Small, Med]
	2	Yes	0.035	0.629	0.734	[Min, PS, Mod, None, Small, Low]
	3	Yes	−0.015	0.619	0.685	[Min, PS, Mod, Relu, Small, Med]
	4	Yes	0.006	0.610	0.677	[Min, PS, Mod, Relu, Small, Med]
	5	No	0.012	0.607	0.656	[Min, PS, Mod, None, Small, Low]

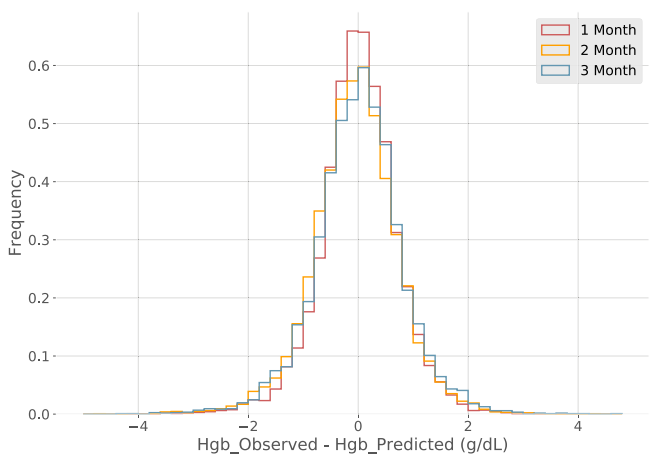

Fig. 9. Distribution of the prediction error (the difference between observed and predicted Hgb levels) for 1, 2, and 3 month predictions for the models that use 5 months of historic data.

than 1 g/dL, and the absolute value of over 95% (93% and 93% respectively) of the prediction errors were less than 1.5 g/dL. These results are typical of the 15 best models listed in Table 2.

- The best performing models were all produced by using the minimum amount of data and 13 of the 15 models were the small version of the model. This is noteworthy because it widens the applicability of this approach in two ways:
  - The information used in the `Min` data models (Hgb, ESA dosing, and iron dosing data) is very likely to be captured by a dialysis organization, and
  - The computational requirements of the models decrease as the size of the model decreases.

Barbieri et al. [14] estimate that the "intra-individual laboratory H[g]b concentration variability is ±0.5 g/dL and that fluctuations of volemia (weight gain and weight loss) may significantly affect the precision of H[g]b measurements"; in light of this it is clear that using even a single month of history produces models whose predictive performance at the 1, 2 and 3 month forecasting horizons are clinically relevant.

5.2. Two examples

The models presented here perform excellently on the stable patients, that is, those patients whose Hgb time series look like the one presented in Fig. 2. This is the least that can be expected from performant models, but the true test is how the models perform on the Hgb time series with cycling and other abrupt and unexpected changes. The two examples that follow are:

1. An example where the performance of the models is excellent, and
2. An example where the performance of the models is not very good.

Fig. 10 provides a specific example of the 1, 2, and 3 month predictions made by the set of models that use 5 months of historic data in which the performance of the models is excellent across the monthly Hgb draws: the MAE of the 1 month predictions shown is 0.222 which is less than 0.527, the MAE of the 1 month forecast horizon model that uses 5 months of historic data; the MAE of the 2 month predictions shown is 0.310 which is less than 0.590, the MAE of the 2 month forecast horizon model that uses 5 months of historic data; and the MAE of the 3 month predictions shown is 0.377, which is less than 0.607, the MAE of the 3 month forecast horizon model that uses 5 months of historic data. Because the models use 5 months of historic data the first 6 monthly Hgb draws do not have any predictions. Of note is the ability of the all three models to forecast a dramatic change in Hgb level even when there is no prior indication of this change, and the combined set of predictions from all three models provide a Hgb level trajectory that indicates how this change will occur. The set of predictions for monthly Hgb draws 7 and 13 are particularly striking examples of this; both sets of predictions precisely capture the Hgb trajectory that is realized over the following three months (monthly Hgb draws 8, 9, and 10, and monthly Hgb draws 14, 15, and 16 respectively). In particular, monthly Hgb draw 7 is the second month in a row where there has been an increase in Hgb level, and the predicted trajectory captures the fact that, although the Hgb level will increase for another month, there is an almost 2 g/dL drop coming over the next 2 months – a quite abrupt change over the trend established during the past 3 months. The set of predictions at monthly Hgb draw 13 again captures the abrupt and unexpected change in trajectory.

If the only information available to the physician or anemia manager is the set of 3 month predictions, at monthly Hgb draw 9 they would anticipate a drop of approximately 0.5 g/dL over the three

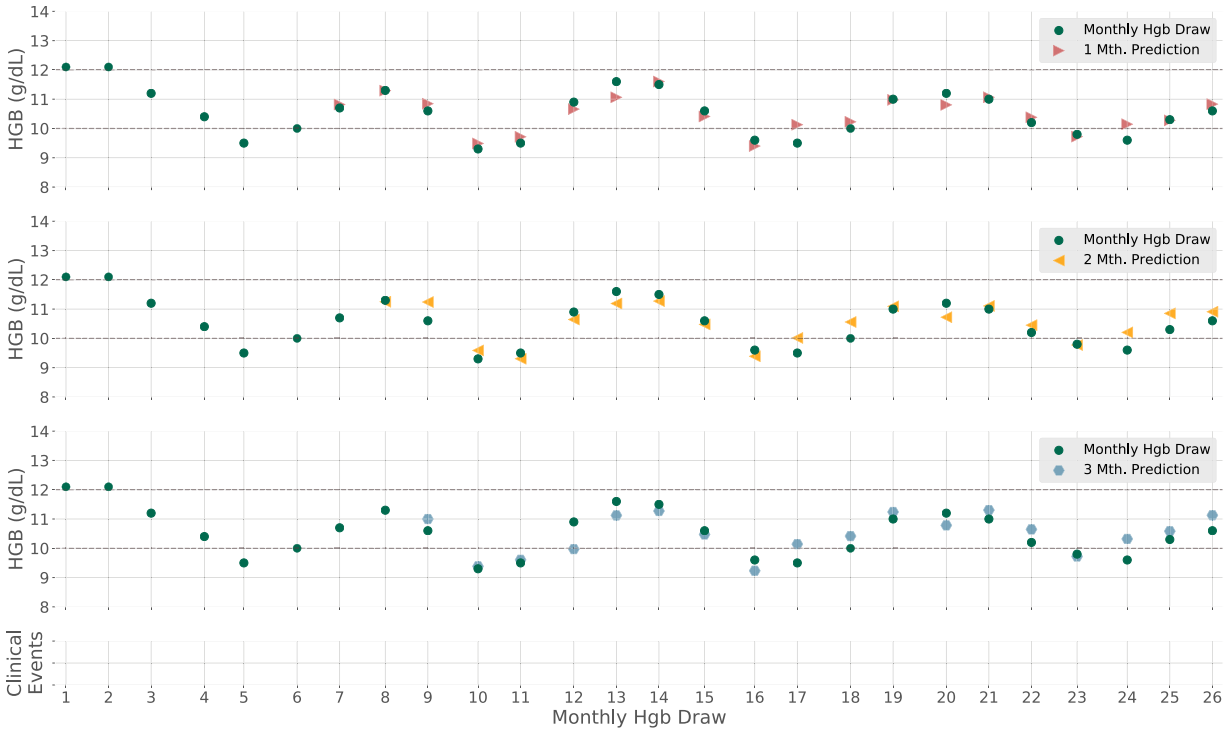

Fig. 10. Plot of the 1 month (top), 2 month (middle), and 3 month (bottom) predictions for the models that use 5 months of historic data where the performance of the models is excellent: the MAE of the 1 month predictions is 0.222, of the 2 month predictions is 0.310, and of the 3 month predictions is 0.377. Clinical events are illustrated on the bottom – hospitalizations as horizontal bars indicating the length of stay and blood transfusions as diamonds indicating the day the transfusion occurred. The data shown in this figure is anonymized patient data.

month period, and would likely be surprised to observe an almost 1.0 g/dL drop at monthly Hgb draw 10. However, the 1 and 2 month predictions made at monthly Hgb draw 9, accurately forecast the unusual trajectory that occurs over the following three months, so what happens at monthly Hgb draws 10 and 11 leading up to monthly Hgb draw 12 is expected and would not be a surprise to the physician or anemia manager. Furthermore, the 1 and 2 month predictions made at monthly Hgb draw 12 are very accurate, so that the anemia manager would, at monthly Hgb draws 10 and 11, have a much better indication of the approximately 1.5 g/dL rise that occurs between monthly Hgb draws 11 and 12.

While Fig. 10 provides an example of the quite excellent performance that the set of models is capable of producing (with errors between the observed and predicted Hgb levels being very small, and the three month Hgb trajectories being captured very accurately), the tails of the distributions shown in Fig. 9 make it clear that performance as good as this cannot always be expected – there are a relatively small number of predictions where the absolute value of the error is more than 2 g/dL. Fig. 11 provides a specific example of the 1, 2, and 3 month predictions made by the set of models that use 5 months of historic data in which the performance of the models is not very good for a number of the monthly Hgb draws: the MAE of the 1 month predictions shown is 0.600 which is greater than 0.527, the MAE of the 1 month forecast horizon model that uses 5 months of historic data; the MAE of the 2 month predictions shown is 0.716 which is greater than 0.590, the MAE of the 2 month forecast horizon model that uses 5 months of historic data; and the MAE of the 3 month predictions shown is 0.592, which is just slightly less than 0.607, the MAE of the 3 month forecast horizon model that uses 5 months of historic data. The predictions for monthly Hgb draws 8, 9, and 15 in Fig. 11 are examples of the 'tails of the distribution' performance, with the difference between the observed and predicted Hgb levels being off by approximately 2 g/dL. This particular patient represents one of the more difficult 'types' of patient to predict – they have hemoglobin cycling as well as 2 very large month-on-month changes (an almost 3 g/dL drop between monthly Hgb draws 7 and 8, and between monthly Hgb draws 21 and 22) that do not have an obvious cause. In addition, the patient has a number of clinical events (3 hospitalizations and 1 transfusion) that occur over the 28 month time period shown, and as such, is missing a number of predictions. For example, because of the transfusion that occurs between monthly Hgb draws 16 and 17, there is no 1 month, 2 month, and 3 month prediction for monthly Hgb draw 17, no 2 month and 3 month prediction for monthly Hgb draw 18, and no 3 month prediction for monthly Hgb draw 19.

However, despite the fact that some predicted Hgb levels fail to match up closely with the observed Hgb levels, by and large the sets of predicted Hgb levels still capture trends (dips and rises in monthly Hgb levels), and would still allow physicians and anemia managers to anticipate such trajectories. Of note is the fact that the set of predictions made at monthly Hgb draw 19 accurately captures the approximately 1 g/dL rise over the following two months as well as the subsequent approximately 3 g/dL drop the following month.

5.3. Discussion

To better understand the contribution of the future iron dosing to the performance of the models, we ran the models *without* the future iron dosing information. Table A.4 of the appendix lists the best performing model for each of the 15 combinations of forecast horizon and number of months of history when *no* future iron dosing information is used. When the performance metrics (ME, MAE, and MSE) for the two best performing models for a given forecast horizon and number of months of history are compared (i.e., comparing the performance metrics on the same row of Table 2 and Table A.4 of the appendix), they are substantially *the same*: the performance metrics generally differ in the second or third decimal place, and are statistically, but more importantly clinically equivalent. This result implies that the models that *do not* use future iron dosing information can be used in place of the

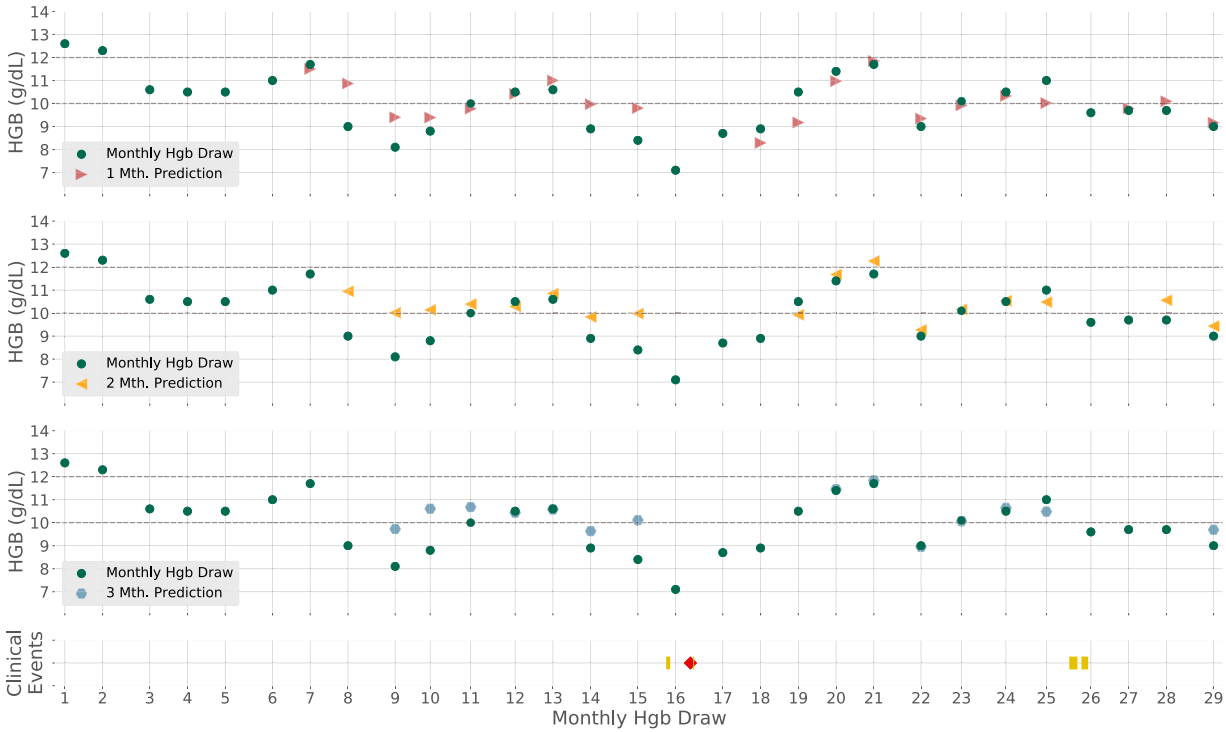

Fig. 11. Plot of the 1 month (top), 2 month (middle), and 3 month (bottom) predictions for the models that use 5 months of historic data where the performance of the models is not very good: the MAE of the 1 month predictions is 0.600, of the 2 month predictions is 0.716, and of the 3 month predictions is 0.592. Clinical events are illustrated on the bottom – hospitalizations as horizontal bars indicating the length of stay and blood transfusions as diamonds indicating the day the transfusion occurred. The data shown in this figure is anonymized patient data.

models that *do* use future iron dosing, with no change in model performance. This is desirable because a simpler model is preferred to a more complicated model (e.g., being able to acquire the information needed to build the model or having the computing resources required), especially if the two models produce equivalent results. Thus in practice it is likely that a model that *does not* include future iron dosing would be used.

The RNN architecture of the models built in this paper specifically accounts for the sequential/time series nature of the data, and the performance of the models indicates that the modeling approach is justified. It is difficult to directly compare the performance of our models with those in the literature: not all models in the literature use the same ESA (EPO), and not all literature uses the same performance metrics (for example while Fuertinger et al. [15] use EPO as the ESA, they do not report any results using ME, MAE, or MSE). Thus any comparisons made here are not direct comparisons, but will provide a rough idea of how the models presented in this work compare to those in the literature. The two comparisons that follow are with models and work that is most similar in nature. Barbieri et al. [13] built a MLP model to predict Hgb levels where the forecasting horizon is 1 month and the ESA used is Darbepoetin; while the forecasting horizon is the same as one of the forecasting horizons used in this paper, the ESA is pharmacokinetically different. Barbieri et al. [13] report an average MAE value (for the difference between the observed Hgb level and the predicted Hgb level) of 0.574 while 0.537 is the average of the MAE values reported in Table 2 for models with a 1 month forecast horizon. Building on the work of Barbieri et al. [13], Barbieri et al. [14] built an ANN model to predict Hgb levels where the forecasting horizon is 3 months and the ESA used is Darbepoetin. Once again while the forecasting horizon is the same as one of the forecasting horizons used in this paper, the ESA is pharmacokinetically different. Barbieri et al. [14] report a MAE value of 0.75 while 0.614 is the average of the MAE values reported in Table 2 for models with a 3 month forecast horizon. In both comparisons the models built in this paper have better (smaller) MAE values for the given forecasting horizons.

The approach developed in this paper is a system which, at any given month, will generate a patient's predicted Hgb trajectory. This trajectory is made up of the 1, 2, and 3 month Hgb level predictions, which are conditioned on the anticipated future ESA dosing pattern (for simplicity we are ignoring iron dosing given the discussion in the first paragraph of this section). Thus, for any given month, it is possible not only to generate the anticipated Hgb trajectory under the future ESA dosing pattern if the ESA dosing protocol is followed, but it is also possible to generate the anticipated Hgb trajectory under other future ESA dosing patterns that do not follow the ESA dosing protocol. This "what if?" analysis can provide the physician or anemia manager with alternative courses of action when making a dosing decision: every alternative (non-protocol) ESA dosing pattern has an expected outcome (i.e., the associated Hgb trajectory), and the physician or anemia manager can consider these alternatives when deciding whether to follow the ESA dosing pattern suggested by the ESA protocol. This consideration is possible precisely because the physician or anemia manager now has

1. The expected Hgb trajectory if they were to follow the ESA dosing pattern prescribed by the protocol, and
2. For each alternative dosing pattern, the expected Hgb trajectory if they were to follow an alternative ESA dosing pattern.

Of course this type of analysis requires the development of a decision support tool which tightly integrates the information produced by this modeling approach with the decision making process, so that the information is provided at the right time and in the right way. Future work should also focus on the best way to deal with the 'tails of the distribution' performance, so that the uncertainty surrounding the predicted Hgb level is also conveyed in the information provided by the decision support tool.

6. Conclusions

This work presented a RNN approach to predicting Hgb levels of End-Stage Renal Disease patients who are on ESAs to address their ESRD-induced anemia. The problem faced by physicians and anemia managers is that the one-size-fits-all protocols that are typically in use do not work for everyone: patient response to the ESA is variable and depends on a large number of factors including inflammation and the presence of raw materials, such as iron, which are needed to make red blood cells. This variable patient response combined with the mismatch in the time between dosing decisions and the time it takes for the effects of the dosing decisions to be realized is a primary contributor to the phenomenon of Hgb cycling. A RNN approach to predicting Hgb levels using historic data together with future ESA and iron dosing data was developed in this paper. To the best of our knowledge this approach is also the first to simultaneously consider both future ESA and iron dosing when predicting Hgb levels. The approach results in a set of 1, 2, and 3 month predictions that provides physicians and anemia managers with a Hgb level trajectory that will likely ensue over the next three months. Results of a large design of experiments indicate that using even a single month of history produces a model whose predictive performance is clinically usable at the 1, 2 and 3 month forecasting horizons.

Further research will focus on trying to find ways to leverage the dialysis and laboratory information – the current results suggest that, as presented to the model, the features based on dialysis metrics and laboratory values do not provide any noticeable predictive lift.

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Declarations of interest

None declared.

Appendix A

Table A.1
The 25 laboratory tests, abbreviations, associated range of acceptable values for inclusion in the model, and units of measurement.

Test name	Abbreviation	Range	Units
Hematocrit	HCT	(10, 60)	%
Sodium	NA	(110, 160)	mmol/L
Carbon dioxide	CO2	(10, 60)	mmol/L
Ferritin	FERITN	(10, 3000)	ng/mL
Int. normalized ratio	PTINR	(0.9, 9)	–
Urea reduction ratio	URR	(40, 100)	%
Red cell distribution width	RDW	(10, 20)	%
Parathyroid hormone intact	PTHINT	(10, 4000)	pg/mL
Phosphorus	PHOS	(1, 20)	mmol/L
Transferrin	TRF	(50, 500)	mg/dL
Creatinine	CREAT	(1, 25)	mg/dL
Iron	IRON	(5, 500)	mcg/dL
Mean corpuscular volume	MCV	(50, 120)	fL
Hemoglobin	HGB	(5, 20)	g/dL
Albumin	ALB	(1, 5)	g/L
Calcium	CALCM	(5, 18)	mg/DL
Sedimentation rate	ESR	(1, 200)	mm/h
Folate, serum	FOLATE	(1, 20)	ng/mL
Hemoglobin A1c	HBA1C	(4.5, 20)	%
Potassium	K	(2, 9)	mmol/L
Kt⧹ V	KTV	(0.5, 3.5)	–
Percent saturation	PSAT	(1, 90)	%
Glomular fill rate	GFRCAL	(1, 35)	mL/min
Protein catabolic rate	PCR	(0.2, 2.0)	g/kg/day
Vancomycin LV	VANCOR	(0, 100)	ug/mL

Table A.2
Number of input samples in each of the three modeling data (sub)sets (Training, Validation, and Testing) for the 15 different combinations of Forecast Horizon and Months of History. The percent of the total samples that belong to each data (sub)set (Training, Validation, and Testing) are shown in parenthesis. The number of data samples is related to the 'Months of History' parameter. The 'Months of History' parameter specifies the number of consecutive monthly Hgb draws there must be in order to generate a sample. The larger the value of this parameter (i.e., the more consecutive monthly Hgb draws there need to be), the smaller the number of samples.

Forecast horizon	Months of history	Number of input samples (% of overall dataset)		
(months)		Training	Validation	Testing
1	1	39938 (69.6%)	9598 (16.7%)	7835 (13.7%)
	2	37152 (67.7%)	9562 (17.4%)	8184 (14.9%)
	3	35905 (68.3%)	9201 (17.5%)	7500 (14.3%)
	4	33940 (67.3%)	8723 (17.3%)	7779 (15.4%)
	5	32414 (67.0%)	8686 (17.9%)	7296 (15.1%)

(*continued on next page*)

Table A.2 (*continued*)

Forecast horizon (months)	Months of history	Number of input samples (% of overall dataset)		
		Training	Validation	Testing
2	1	35630 (69.7%)	8514 (16.7%)	6954 (13.6%)
	2	33120 (67.6%)	8561 (17.5%)	7291 (14.9%)
	3	32123 (68.4%)	8152 (17.4%)	6701 (14.3%)
	4	30260 (67.1%)	7856 (17.4%)	6965 (15.4%)
	5	28909 (66.8%)	7832 (18.1%)	6524 (15.1%)
3	1	32168 (69.8%)	7657 (16.6%)	6242 (13.5%)
	2	29831 (67.5%)	7754 (17.5%)	6602 (14.9%)
	3	29053 (68.5%)	7296 (17.2%)	6049 (14.3%)
	4	27256 (67.0%)	7165 (17.6%)	6280 (15.4%)
	5	26066 (66.7%)	7135 (18.3%)	5887 (15.1%)

Table A.3
Performance statistics for the training, validation, and testing sets of the best performing network configuration for each of the 15 different combinations of forecast horizon and months of history when future iron dosing is used. The best performing network configuration for a given combination of forecast horizon and months of history is the network configuration that produces the minimum MAE value when used on the testing set and can be found in Table 2.

Forecast horizon (months)	Months of history	Training set			Validation set			Testing set		
		ME	MAE	MSE	ME	MAE	MSE	ME	MAE	MSE
1	1	−0.098	0.530	0.499	−0.081	0.565	0.558	0.010	0.541	0.521
	2	−0.030	0.514	0.470	−0.013	0.527	0.483	−0.008	0.548	0.535
	3	0.029	0.489	0.428	0.036	0.532	0.492	−0.004	0.537	0.513
	4	−0.101	0.507	0.462	−0.092	0.497	0.434	0.010	0.534	0.508
	5	0.029	0.482	0.416	0.034	0.522	0.487	0.017	0.527	0.489
2	1	−0.083	0.581	0.601	−0.078	0.634	0.700	0.009	0.600	0.635
	2	−0.085	0.534	0.523	−0.072	0.573	0.569	0.027	0.608	0.669
	3	−0.036	0.546	0.537	−0.031	0.578	0.585	−0.046	0.605	0.664
	4	0.008	0.541	0.529	0.011	0.556	0.543	0.005	0.603	0.649
	5	−0.066	0.526	0.501	−0.051	0.577	0.598	−0.047	0.590	0.615
3	1	−0.031	0.586	0.618	−0.035	0.644	0.734	−0.033	0.605	0.658
	2	0.022	0.577	0.601	0.044	0.591	0.606	0.035	0.629	0.734
	3	0.006	0.553	0.554	−0.003	0.596	0.625	−0.015	0.619	0.685
	4	0.049	0.539	0.542	0.051	0.579	0.592	0.006	0.610	0.677
	5	−0.009	0.554	0.555	0.006	0.594	0.645	0.012	0.607	0.656

Table A.4
The best performing network configuration for each of the 15 different combinations of forecast horizon and months of history when *no* future iron dosing information is used. The best performing network configuration for a given combination of forecast horizon and months of history is the network configuration that produces the minimum MAE value when used on the testing set.

Forecast horizon (months)	Months of history	Model modified?	Testing set			Network Configuration
			ME	MAE	MSE	
1	1	Yes	0.018	0.541	0.521	[Min, PS, Orig, Relu, Small, Low]
	2	No	0.017	0.550	0.531	[Min, PS, Mod, None, Small, Low]
	3	Yes	−0.001	0.539	0.516	[Min, PS, Mod, None, Small, Med]
	4	Yes	0.015	0.537	0.513	[Min, PS, Mod, Relu, Small, Low]
	5	Yes	0.017	0.530	0.488	[Min, WPS, Mod, Relu, Small, Low]
2	1	No	−0.031	0.601	0.638	[Min, PS, Orig, None, Small, Med]
	2	No	−0.005	0.618	0.686	[Min, WPS, Mod, None, Small, Low]
	3	Yes	−0.008	0.607	0.657	[Min, WPS, Mod, Relu, Small, Low]
	4	No	−0.014	0.607	0.668	[Min, WPS, Mod, Relu, Small, Med]
	5	Yes	0.018	0.595	0.618	[Min, PS, Mod, None, Small, Med]
3	1	Yes	0.000	0.608	0.656	[Min, PS, Orig, Relu, Small, Med]
	2	Yes	0.029	0.634	0.734	[Min, PS, Mod, None, Small, Low]
	3	Yes	−0.021	0.622	0.693	[Min, PS, Mod, Relu, Small, Low]
	4	Yes	0.003	0.619	0.698	[Min, PS, Mod, Relu, Small, Med]
	5	Yes	0.012	0.606	0.651	[Min, WPS, Mod, None, Small, Low]

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